

Department of Mechanical Engineering
Mid Semester Test I, September 2025
Fundamentals of Mechanical Engineering

Duration: 1:00 Hrs,
All questions are compulsory

Max Marks:20

Q.1 (i) A material has modulus of rigidity equal to $0.4 \times 10^5 \text{ N/mm}^2$ and bulk modulus equal to $0.8 \times 10^5 \text{ N/mm}^2$. Find its Young's Modulus and Poisson's Ratio.
[CO 1, Marks 5]

(ii) A steel rod 400 cms. long and 20 mm. diameter is subjected to an axial tensile load of 4500 kg. Find the change in length and induced stress Take $E_s = 2 \times 10^6 \text{ kg./cm}^2$. [CO 1, Marks 5]

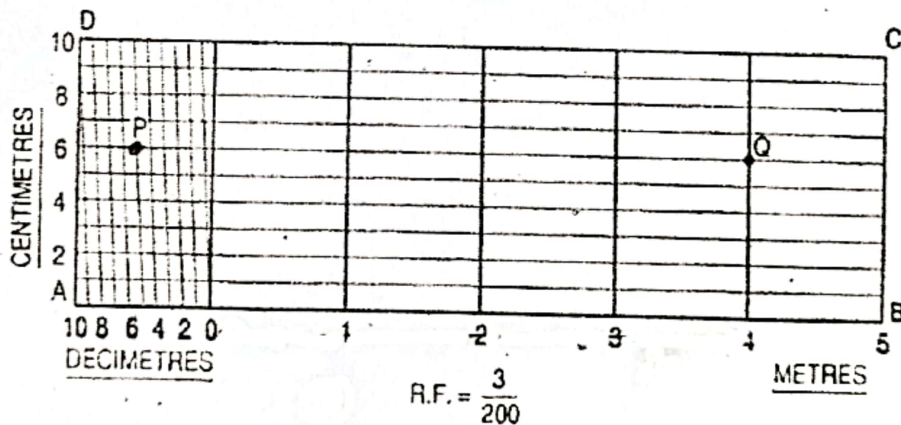
OR

(i) Define any five properties of engineering materials.
[CO 1, Marks 5]

(ii) Draw and explain Stress and strain Curve for a ductile material.
[CO 1, Marks 5]

Q.2 (i) A distance of 6 m is represented by 2 cm on a drawing. If a scale is to be prepared for measuring a distance up to 45 m, determine RF and length of scale.
[CO 5, Marks 3]

(ii) Read the distance between P and Q on the scale as shown in figure below and name the scale. [CO 5, Marks 2]



(iii) Draw neatly symbols for first and third angle projection.
[CO 5, Marks 2]

(iv) Classify Engineering Materials.
[CO 5, Marks 3]
